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PLAIN-ENGLISH SAFETY REVIEW · v0.6.4

# Bitcoin Easy Signer

Is this software safe to use? Three independent reviews, one day, one plain answer — with the full audit trail in everyday language: what was found, how serious it really was, and what became of it. Written for the spouse, the trustee, the lawyer, the accountant.

Prepared by Z.ai from the public Color Team  
reviews · 2 October 2026

# Bitcoin Easy Signer

Plain-English Safety Review · Version 0.6.4 · 2 October 2026

## GREEN

THE HIGHEST GRADE THIS REVIEW CAN GIVE — EARNED, NOT GIVEN

This review was written for the person this software is actually for: a spouse, a trustee, a lawyer, an accountant, a family helper. It answers the questions such a person asks, in ordinary language. Every statement in these pages is drawn from — and can be checked against — three full technical reviews completed on 2 October 2026 by an independent audit panel run by Z.ai. Those reviews, their evidence, and every checksum are public; the last page of this document tells you exactly where.

### Can this app steal my Bitcoin?

**No — and no way to do it was found, in any of the three reviews.** The reviewers actively tried: one examiner whose only job was to attack the software looked for any path to move money without the owner's approval, and every attempt failed. There is also a protection you can verify yourself, without trusting anyone: **the app never holds your keys.** It has no way to read your seed phrase or passwords, and it cannot spend anything unless you physically approve the payment on the screen of your hardware wallet — the little device itself, not the computer.

### Could it give away my keys or passwords?

**No.** The software contains no code that can read, store, or send a seed phrase, private key, or device PIN. The reviewers traced every path by which information enters or leaves the app; the only things it ever writes to your computer are an unsigned payment file and a troubleshooting report that contains no wallet details.

### Could someone change a payment without me seeing it?

**No way was found.** The app rebuilds and re-checks the payment against what you approved on screen — several times, in several independent ways — before anything is sent, and again after. If the payment that comes back from your devices differed from the one you approved by even one character, the app refuses it. The reviewers tested this by tampering with payments deliberately; the tampered versions were all rejected.

### What does “Green” mean — and not mean?

It means every category of the review passed, under grade rules that were written down and locked *before* the review began, so they could not be bent to produce a nicer result — the same rules that, hours earlier, gave this software its second-highest grade rather than look the other way. It does **not** mean a guarantee, and it does not cover future versions: it is a clean bill for **version 0.6.4, examined on 2 October 2026**. Your own eyes still matter most: check the address and amount on your hardware

wallet's screen every time, as the manual says.

**Who did this review, and can I check it?**

An independent review panel run by Z.ai, using a team of examiner programs — an attacker, a defender, a mathematics specialist, a hardware specialist, a supply inspector, and a referee whose only job is to re-check the others' work. Nothing appears in this document that the referee could not personally verify. You do not have to take our word for any of it: **every page of the technical evidence is public** — you can read our homework yourself, starting from the last page of this document.

## How This Review Was Done

Three separate reviews were completed in a single day, each on the exact published version it covers. Each review follows the same rules: the grading scale is locked in writing before anyone looks at the code; six independent examiners do the work, none seeing the others' findings until the end; a referee re-checks every important claim personally; and the final grade is the *lowest* result on the panel — one bad finding fails the whole review, no matter how good everything else is.

The three reviews, in one sentence each:

|          |                                                                                                                                                                                                                                                                                 |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Review 1 | The full examination. 25 observations recorded — <b>none of them a way to steal money or keys</b> — each with a written repair. Version examined: 0.6.2.                                                                                                                        |
| Review 2 | The re-check. Every repair verified in place — and locked there by automatic alarms that go off if anyone ever undoes one. One weakness remained, in how new versions are <i>released</i> , not in the software itself. Grade: one step below the top. Version examined: 0.6.3. |
| Review 3 | The final check. The release process now runs itself, with tamper-checks that executed, in order, before this version was published — and the version you download is provably the exact one that passed every test. Grade: <b>Green</b> . Version examined: 0.6.4.             |

The pages that follow are the audit trail: what was found across all three reviews, in plain language, each item labeled by how serious it actually was, and what became of it.

## The Review Team

Six independent examiners did this work. Each has exactly one job, none sees the others' findings until the end, and the final grade is the *lowest* score on the team — one bad finding fails the whole review, no matter how good everything else is.

|                               |                                                                                                                                                              |                                                                                         |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| RED · The attacker            | Try to steal the money, the keys, or the owner's approval — by any trick: forged inputs, swapped devices, lying network services, tampered components.       | No way in was found, in any of the three reviews.                                       |
| BLUE · The defender           | Confirm every promised protection is real, works end to end, and is watched by an automatic alarm that fails the software if anyone ever removes it.         | Every protection verified; every repair from earlier reviews confirmed locked in place. |
| ORANGE · The logic specialist | Check the mathematics and money-handling the app cannot afford to get wrong — including the free public building blocks it stands on, examined line by line. | All money logic proven correct against Bitcoin's official test cases.                   |

|                                         |                                                                                                                                                                                                             |                                                                                          |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| <b>COPPER · The hardware specialist</b> | Check everything between the app and the little signing devices — the connections, the drivers, and whether a fake device could deceive the app.                                                            | Connections verified; a fake can consume an approval attempt but cannot forge a payment. |
| <b>AMBER · The supply inspector</b>     | Check how the app is born: every component it is built from, the machinery that builds and packages it, and the delivery to your computer — so nothing can be tampered with between the developers and you. | The chain held; the version you download is provably the one that passed every test.     |
| <b>WHITE · The referee</b>              | Distrust everyone else's work. Re-check every important claim personally, and refuse to publish anything that cannot be verified. Only the referee can approve the final grade.                             | Every load-bearing claim re-derived and confirmed; the grade is the referee's ruling.    |

The team is assembled and its rules locked before anyone looks at the code — so the grade cannot be negotiated afterward. The same rules gave this software its second-highest grade earlier the same day, when one weakness remained; the Green came only after that weakness was fixed and demonstrated.

## The Audit Trail

**Read this first:** across all three reviews, no way was ever found for this software — or the components it is built on — to move Bitcoin without the owner's approval, or to expose keys or passwords. The items below are about manufacturing discipline (how the software is built, packaged, and released) and extra safety margins. The labels mean:

|                          |                                                                                            |
|--------------------------|--------------------------------------------------------------------------------------------|
| <b>DANGER SIGN</b>       | Could put money or secrets at risk. <b>None were found — in any review.</b>                |
| <b>IMPORTANT TO FIX</b>  | A weakness in how the software is built or released — not in what it does when you use it. |
| <b>MINOR IMPROVEMENT</b> | Extra safety margin or thoroughness; nothing a user could be harmed by.                    |
| <b>HOUSEKEEPING</b>      | Notes for the maintainers; no effect on users.                                             |

### From Review 1 — the full examination

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                           |                   |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>IMPORTANT TO FIX — RESOLVED</b> | <b>A tool used while preparing releases was downloaded from the internet without a tamper-check.</b> It never touched anyone's Bitcoin and did not affect the released app — but a future release could theoretically have been tampered with this way. Fixed: every component is now verified against a tamper-proof record before it is used, and the check runs before anything sensitive is unlocked. | Report 1, item 01 |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                          |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| <b>IMPORTANT TO FIX — RESOLVED</b>  | <b>Weaknesses in the release machinery.</b> Several steps relied on a person doing the right thing in the right order rather than on the machine enforcing it. Not user-facing and never exploitable against a released version — but process is protection. Fixed in two rounds: the release pipeline was rebuilt so the tamper-checks run themselves, and the final review watched them do exactly that.                                                                                                                                                                                             | Report 1, items 03-05; Report 2, item 33 |
| <b>IMPORTANT TO FIX — RESOLVED</b>  | <b>The USB software could load from the wrong place.</b> The component that talks to hardware wallets could, on some Macs, pick up a stray copy of itself instead of the one shipped with the app. Fixed: the correct copy is now locked in by name, verified at startup, and the app refuses to run the device step without it.                                                                                                                                                                                                                                                                       | Report 1, items 07, 11-12                |
| <b>MINOR IMPROVEMENT — RESOLVED</b> | <b>The building blocks had never been independently checked.</b> Like most software, this app stands on free, public components written by others. The reviewers examined them line by line and found small flaws — none reachable in this app — including one bug the original authors have not fixed to this day. The team's response became the proudest item on this page: exact copies are now locked inside the project, every line reviewed, and that upstream bug is fixed here first. This app now runs on a version of its own foundation that is, in one respect, better than the original. | Reports 1-2, items 02, 09-17, 38         |
| <b>MINOR IMPROVEMENT — RESOLVED</b> | Assorted hardening: better failure messages, more complete records of what ships in each version, and similar polish.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Report 1, items 18-21                    |

## From Review 2 — the re-check

Every item above was verified fixed — and, importantly, each repair is pinned by an automatic test that fails loudly if anyone ever undoes it. The reviewers checked this the hard way: they deliberately undid each repair in a scratch copy and confirmed the alarm went off. One new item was recorded:

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                   |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>IMPORTANT TO FIX — RESOLVED</b> | <b>The release was published by hand instead of by the new machine gates.</b> The rebuilt release pipeline existed and its checks were real — but for this one release, a person performed the final publishing step manually, so the machine's checks never actually ran. The published files were verified identical to the tested ones, so no harm was possible — but the review's rules are applied as written: a stated safety check that did not run is a finding. This single item was the only thing standing between the software and its top grade. | Report 2, item 33 |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|

## From Review 3 — the final check (this version, 0.6.4)

The response to that finding was stronger than the review required: manual publishing is now outright prohibited (“if the machinery is unavailable, wait”), and this version was then published **by the machinery itself**. The referee verified, from the machine's own logs: the tamper-checks ran, in order, before the release was created, and the version you download is provably the exact one that passed every

test. Two rehearsal attempts that misfired earlier were correctly refused by the machinery — evidence the checks are real. The items recorded in this final round, all minor:

|                   |                                                                                                                                                                                                                                       |                                  |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
|                   |                                                                                                                                                                                                                                       |                                  |
| MINOR IMPROVEMENT | The new “no manual publishing” policy has not yet been countersigned and dated by the owner. The grade does not rest on the policy — it rests on the machinery demonstrably doing the work — but a signature is the tidy final touch. | Report 3, item 45                |
| MINOR IMPROVEMENT | A few hardening notes carried forward by choice (a seconds-wide substitution window that requires physical access to your own computer; deeper record-keeping). Each is documented with its reason; none can affect a payment.        | Reports 2-3, items 26, 29-32, 46 |
| HOUSEKEEPING      | Bookkeeping and style notes: how files are listed, a timestamp quirk in GitHub's records, and similar. Listed for completeness in the technical reports.                                                                              | Report 3, items 47-53            |

## What This Review Does Not Cover

- It covers version 0.6.4 exactly as published on 2 October 2026. New versions need a fresh review; this document should not be read as covering them.
- The final round did not repeat the physical-device test — it was not needed, because this version changed no application behavior. The owner's own successful device test, on the prior version, is recorded in the project's release notes.
- The review was performed by an AI audit panel. It is thorough and its evidence is public, but it complements — and does not replace — a qualified human security firm, which this project has not retained.
- “Nothing found” always means “nothing found within this review's coverage,” never “nothing exists.”

## Check Our Homework

Everything summarized here exists in full detail, publicly:

|                                 |                                                                                    |                                                                |
|---------------------------------|------------------------------------------------------------------------------------|----------------------------------------------------------------|
|                                 |                                                                                    |                                                                |
| Technical report, version 0.6.2 | The full examination: all 25 items with evidence and repairs.                      | bitcoin-easy-multisig-signer on GitHub; this site, Audits page |
| Technical report, version 0.6.3 | The re-check: every repair verified, and the one finding that held the grade back. | Same locations                                                 |

|                                 |                                                                                                     |                                                                                                                     |
|---------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Technical report, version 0.6.4 | The final check and the grade evidence, from the machine's own logs.                                | Same locations                                                                                                      |
| The audit method itself         | The published rules the panel works under — including the grading scale, locked before each review. | <a href="https://github.com/cjtsh/ai-color-team-audit-framework">github.com/cjtsh/ai-color-team-audit-framework</a> |

*Prepared 2 October 2026 by Z.ai from the three public reviews of Bitcoin Easy Signer versions 0.6.2, 0.6.3, and 0.6.4. Every statement above traces to those reports. No wallet details, addresses, or transaction information appear in this document. A safety review is evidence about one version on one day — it is not a guarantee, and the protections that matter most are the ones you can see yourself: your keys stay on your devices, and you approve every payment on the device screen.*